

FIITJEE

ALL INDIA TEST SERIES

FULL TEST – II

JEE (Main)-2022

TEST DATE: 25-12-2021

Time Allotted: 3 Hours

Maximum Marks: 300

General Instructions:

- The test consists of total 90 questions.
- Each subject (PCM) has 30 questions.
- This question paper contains **Three Parts**.
- **Part-A** is Physics, **Part-B** is Chemistry and **Part-C** is Mathematics.
- Each part has only two sections: **Section-A** and **Section-B**.
- **Section – A** : Attempt all questions.
- **Section – B** : Do any five questions out of 10 Questions.

Section-A (01 – 20, 31 – 50, 61 – 80) contains 60 multiple choice questions which have **only one correct answer**. Each question carries **+4 marks** for correct answer and **-1 mark** for wrong answer.

Section-B (21 – 30, 51 – 60, 81 – 90) contains 30 Numerical answer type questions with answer XXXXX.XX and each question carries **+4 marks** for correct answer. There is no negative marking.

Physics

PART – A

SECTION – A (One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

1. A cannon fires successively two shells with speed 10 m/s, the first at an angle 60° and the second at an angle 45° to the horizontal in same plane. Find horizontal distance where firing leads to collision of shells (take $g = 10 \text{ m/s}^2$ and $\sqrt{3} = 1.7$)

- (A) 3 m
(B) 5 m
(C) 7 m
(D) 9 m

Ans. C

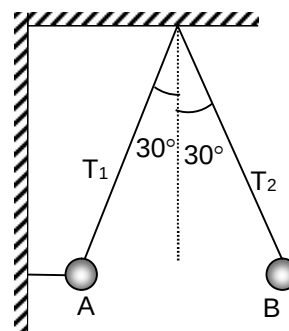
Sol. At the point of collision

$$x \tan 60^\circ - \frac{1}{2} \frac{gx^2}{u^2 \cos^2 60^\circ} = x \tan 45^\circ - \frac{1}{2} \frac{gx^2}{u^2 \cos^2 45^\circ}$$

$$\Rightarrow \frac{gx}{2u^2} (4 - 2) = \sqrt{3} - 1$$

$$\Rightarrow x = \frac{u^2}{g} (\sqrt{3} - 1) = 7 \text{ m}$$

2. A ball is held at rest in position A and another identical ball is released from rest in position B. The ratio of tension T_1 and T_2 is: (string connecting wall and ball A is horizontal)



- (A) 1
(B) $\frac{1}{2}$
(C) $\frac{4}{3}$
(D) $\frac{3}{4}$

Ans. C

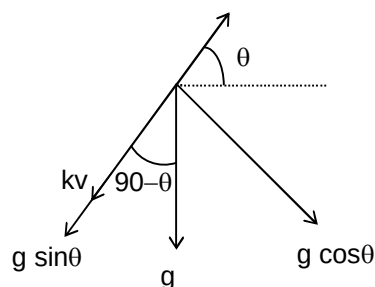
Sol. $T_1 \cos 30^\circ = mg$
 and $T_2 = mg \cos 30^\circ$
 $\frac{T_1}{T_2} = \sec^2 30^\circ = \frac{4}{3}$

3. If there is air drag in the ascent of a ground to ground projectile:

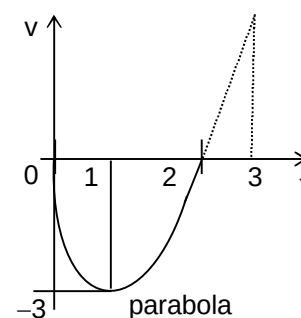
- (A) The magnitude of tangential acceleration increases while magnitude of normal acceleration decreases.
- (B) The magnitude of tangential acceleration decreases while magnitude of normal acceleration increases
- (C) Both increase
- (D) Both decrease

Ans. B

Sol. Tangential acceleration = $-g \sin \theta - kv$
 Normal acceleration = $g \cos \theta$



4. A particle moves along x-axis. Its velocity time graph is shown in figure. The distance travelled by particle in 0 to 3s.



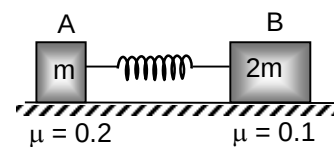
- (A) zero
- (B) 4m
- (C) 8m
- (D) 12m

Ans. C

Sol. Equation of parabola
 $v = 3t^2 - 6t$

$$\text{Distance} = \left| \int_0^2 v dt \right| + \left| \int_2^3 v dt \right| = 8\text{m}$$

5. In the given figure mass of block A is m and that B is $2m$. They are attached at the ends of spring and kept on horizontal surface. Now the spring is compressed by x_0 and released. Find displacement of block B by the time compression of the spring is reduced to $\frac{x_0}{2}$. Assume both are slipping in opposite directions together.



- (A) $\frac{x_0}{3}$
 (B) $\frac{x_0}{2}$
 (C) $\frac{x_0}{6}$
 (D) $\frac{x_0}{4}$

Ans. C

Sol. Since net force on the system of two blocks is zero.

$$x_A + x_B = \frac{x_0}{2}$$

$$mx_A = 2mx_B$$

$$\Rightarrow x_B = \frac{x_0}{6}$$

6. A pendulum bob of mass m with string length ℓ can swing in a vertical plane. The bob is given an initial velocity from lowermost point in horizontal direction such that tension in rope and speed of pendulum becomes zero simultaneously. Find maximum height rise by pendulum.

- (A) $\frac{\ell}{2}$
 (B) $\frac{\ell}{3}$
 (C) ℓ
 (D) $\frac{4\ell}{3}$

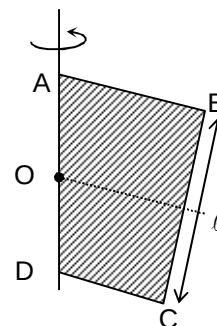
Ans. C

Sol. Given case is only possible for

$$u = \sqrt{2g\ell}$$

So, maximum height = ℓ

7. A sheet ABCD has mass M . Sides AB and CD are parallel to each other and perpendicular to BC. The perpendicular distance of mid point of AD from side BC is half of length of BC then moment of inertia of sheet about axis AD is

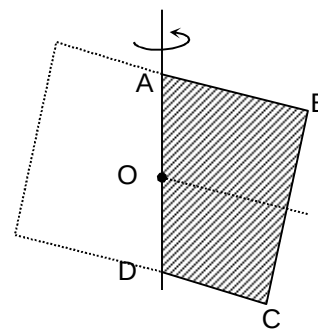


- (A) $\frac{M\ell^2}{3}$
 (B) $\frac{M\ell^2}{2}$
 (C) $\frac{M\ell^2}{12}$
 (D) $\frac{4M\ell^2}{3}$

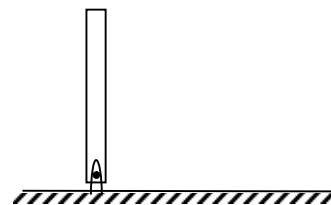
Ans. C

Sol. Plate is half of square plate

$$\text{So, } I = \frac{M\ell^2}{12}$$



8. A uniform rod of m mass is hinged at and kept one end at initially vertical at rest. It is disturbed slightly. The hinge force when rod is about to hit ground.



- (A) $\frac{3}{2}mg$

- (B) $\frac{1}{4}mg$
- (C) $\frac{\sqrt{37}}{4}mg$
- (D) $\frac{\sqrt{41}}{4}mg$

Ans. C

Sol. Final angular velocity (MEC)

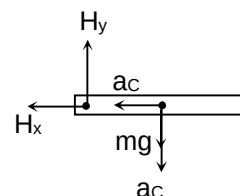
$$\frac{1}{2} \cdot \frac{m\ell^2}{3} \omega^2 = Mg \frac{\ell}{2}$$

$$\Rightarrow \omega^2 \frac{\ell}{2} = \frac{3g}{2}$$

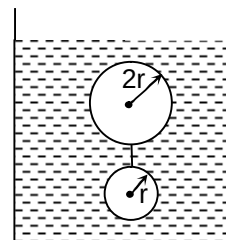
Angular acceleration

$$Mg \frac{\ell}{2} = \frac{m\ell^2}{3} \alpha \Rightarrow \alpha \frac{\ell}{2} = \frac{3g}{4}$$

$$H_x = \frac{3mg}{2}, H_y = \frac{mg}{4} \Rightarrow H = \frac{\sqrt{37}}{4}mg$$



9. Two spherical balls of radius $2r$ and r both having density σ are tied up with light string and released in a viscous fluid of density $\rho > \sigma$ with the rope just taut. The tension in rope when spheres attain terminal velocity:



- (A) $\frac{3}{5}\pi r^3(\rho - \sigma)g$
- (B) $\frac{8}{3}\pi r^3(\rho - \sigma)g$
- (C) $\frac{10}{5}\pi r^3(\rho - \sigma)g$
- (D) $\frac{5}{3}\pi r^3(\rho - \sigma)g$

Ans. B

Sol. At internal velocity Balancing forces

$$Mg \left(\frac{\rho}{\sigma} - 1 \right) = T + 6\pi(2r)\eta v$$

$$mg\left(\frac{\rho}{\sigma} - 1\right) = 6\pi r\eta v - T$$

$$Mg\left(\frac{\rho}{\sigma} - 1\right) = T + 2\left[mg\left(\frac{\rho}{\sigma} - 1\right) + T\right]$$

$$T = \left(\frac{M - 2m}{3}\right)\left(\frac{\rho}{\sigma} - 1\right)g = \frac{8}{3}\pi r^3(\rho - \sigma)g$$

10. A planet of mass m revolves around sun of (mass M) along an elliptical orbit such that its minimum and maximum distance from centre of the sun are r and $3r$ respectively. Its time period of revolution will be

(A) $4\pi\sqrt{\frac{2r^3}{GM}}$

(B) $2\pi\sqrt{\frac{r^3}{GM}}$

(C) $\pi\sqrt{\frac{2r^3}{GM}}$

(D) $\frac{\pi}{2}\sqrt{\frac{r^3}{GM}}$

Ans. A

Sol. $a + ae = 3r$
 $a - ae = r$
 $\Rightarrow a = 2r$
 $T = \frac{2\pi}{\sqrt{GM}} a^{3/2}$

11. Potential energy of a particle of mass 2kg is given as

$$\frac{x^3}{6} - 2x$$

Initially particle is at equilibrium position, then time period for small oscillation about stable equilibrium position is

(A) $2\pi \text{ sec}$

(B) $\frac{\pi}{2} \text{ sec}$

(C) $\frac{4\pi}{3} \text{ sec}$

(D) $\frac{\pi}{3} \text{ sec}$

Ans. A

Sol. At stable equilibrium

$$dF = -2dx$$

$$k = 2$$

$$T = 2\pi\sqrt{\frac{2}{2}} = 2\pi \text{ sec}$$

12. Two moles of an ideal monoatomic gas undergoes into a process such that internal energy depends on entropy $S = \ell \log_e U$. Molar specific heat of gas is

(A) 3

(B) 1

(C) 0.5

(D) 5

Ans. C

Sol. $\frac{dU}{U} = dS$

$$\frac{dT}{T} = \frac{dQ}{T}$$

$$ncdT = dT$$

$$c = \frac{1}{n}$$

13. A point source is placed at 1 m perpendicular height of a square plate of side 2m from its center. A source is producing sound with power $18 \mu\text{J}$. The amount of energy collected by plate per second.

(A) $2\mu\text{J}$

(B) $3 \mu\text{J}$

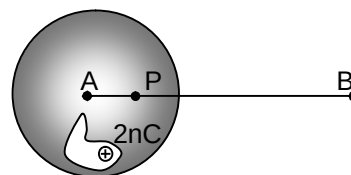
(C) 0.18 J

(D) 0.1J

Ans. B

Sol. Similar to Gauss's Law

14. A point charge 1 nC is placed at 50 cm from the centre of a conducting shell of radius 20 cm having -3 nC charge. There is a cavity in the solid sphere having a positive charge 2 nC . Due to system of shell potential at point P ($AP = 10 \text{ cm}$) as shown in the figure, is



(A) -27 V

(B) -45 V

(C) -49.5 V

(D) -22.5 V

Ans. C

Sol. Potential of shell due to spherical system at P = $= 9 \times 10^9 \times 10^{-9} \left(\frac{1}{0.2} - \frac{1}{0.1} - \frac{1}{0.4} \right) = -49.5 \text{ V}$

Which is also potential of P.

15. In measurement of resistance by meter bridge. The resistivity of wire increases linearly. Then, the percentage error is minimum when null point is at

(A) close to 50 cm

(B) close to 57 cm

(C) close to 45 cm

(D) close to 83 cm

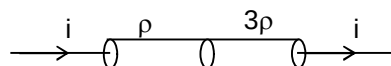
Ans. B

Sol. $Q = P \left(\frac{x^2}{100^2 - x^2} \right)$

$$\frac{dQ}{Q} = \frac{2dx}{x} + \frac{2xdx}{100^2 - x^2}$$

For minimum $\frac{dQ}{Q} \Rightarrow x \rightarrow 57 \text{ cm}$

16. Two conductors of same length and area of cross section are connected in series as shown in the figure. An electric current i flows through them. The total charge accumulated at junction would be (ρ and 2ρ are their resistivity)



(A) $i\rho\epsilon_0$

(B) $2i\rho\epsilon_0$

(C) $\sqrt{3} i\rho\epsilon_0$

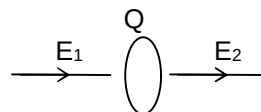
(D) zero

Ans. B

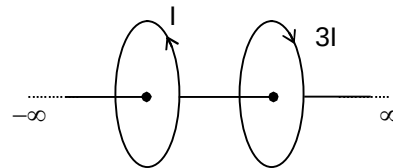
Sol. $E_1 = \rho J$
 $E_2 = 3\rho J$

$$\Delta\phi = \frac{Q}{\epsilon_0}$$

$$\Rightarrow Q = 2i\rho\epsilon_0$$



17. Two co-axial current carrying loops having currents in opposite directions. Then value of $\int_{-\infty}^{\infty} B \cdot dx$, where B is magnetic field on x-axis will be



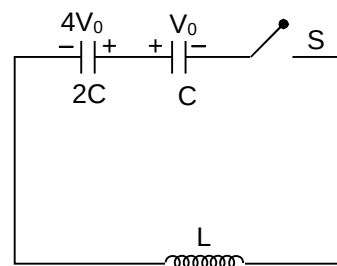
- (A) $\mu_0 I$
 (B) $2\mu_0 I$
 (C) $-2\mu_0 I$
 (D) $-\mu_0 I$

Ans. B

Sol. For one ring $\int_{-\infty}^{\infty} B \cdot dx = \mu_0 I_0$

Due to opposite direction field will cancel out.

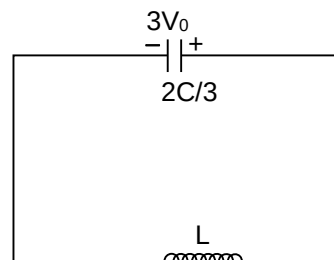
18. In the arrangement shown the maximum current in circuit when switch is closed.



- (A) $V_0 \sqrt{\frac{6C}{L}}$
 (B) $V_0 \sqrt{\frac{C}{L}}$
 (C) $\frac{V_0}{2} \sqrt{\frac{C}{6L}}$
 (D) $\frac{V_0}{2} \sqrt{\frac{6C}{L}}$

Ans. A

Sol. maximum current = $\frac{Q_{\max}}{\sqrt{LC}} = \frac{2CV_0}{\sqrt{L \cdot \frac{2C}{3}}}$
 $= V_0 \sqrt{\frac{6C}{L}}$



19. A toroid has mean radius $R = \frac{20}{\pi}$ cm and a total of 400 turns of wire carrying a current of 2A. A metallic ring at 300 K inside the toroid provides core. If the magnetization intensity $I = 4.8 \times 10^{-2}$ A/m the susceptibility of metal is:

- (A) 2.4×10^{-5}
 (B) 4.2×10^{-5}
 (C) 2.4×10^{-4}
 (D) 4.2×10^{-4}

Ans. A

Sol. $\chi_{cm} = \frac{I}{H} = \frac{I}{ni} = \frac{4.8 \times 10^{-2}}{10^3 \times 2} = 2.4 \times 10^{-5}$

20. An EM wave enters from vacuum into a non magnetic medium of refractive index 1.5. Then the ratio of electric fields in vacuum and medium will be

- (A) $\frac{3}{2}$
 (B) $\frac{2}{3}$
 (C) $\frac{1}{2}$
 (D) 2

Ans. A

Sol. $E_0 = cB_0$
 $E = vB$
 $\frac{E_0}{E} = \frac{c}{v} = n$ {B = B₀ due to nonmagnetic medium}
 $= 1.5$

SECTION – B (Numerical Answer Type)

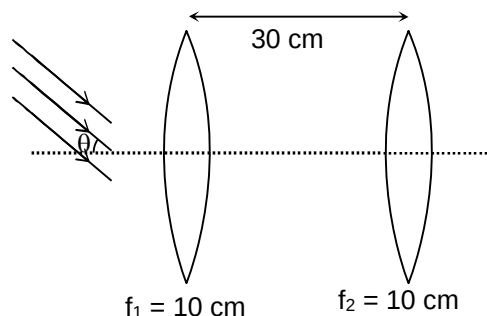
This section contains **10** questions. The answer to each question is a **NUMERICAL VALUE**. For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

21. An ideal choke takes a current of 8A when connected to an AC source of 100 V and 50 Hz. A pure resistance under the same conditions takes a current of 10A. If two are connected in series to an AC supply of 100 V and 40 Hz then current in the circuit is $k\sqrt{2}$. Find the value of k.

Ans. 00005.00

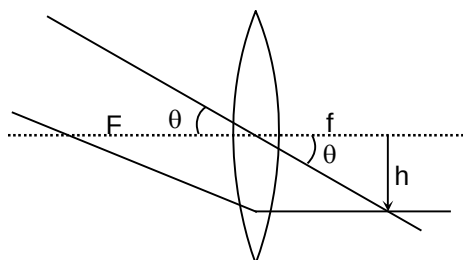
Sol. $X_L = \frac{100}{8} = 12.5\Omega \Rightarrow L = \frac{12.5}{100\pi}$
 $R = \frac{100}{10} = 10\Omega$
 at 40 Hz
 $X_L = 10\Omega \Rightarrow Z = \sqrt{10^2 + 10^2} = 10\sqrt{2}\Omega$
 $i = \frac{V}{Z} = 5\sqrt{2} \text{ A}$

22. A combination of two lenses is used to converge parallel beam. If final image is at 1 cm height from axis, then value of $\tan \theta$ is

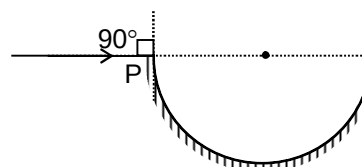


Ans. 00000.10

Sol. $h = f \tan \theta$
 $h = 10 \tan \theta$
 final image has 1 cm height
 So, $h = 1 \text{ cm}$
 Hence $\tan \theta = 0.1$

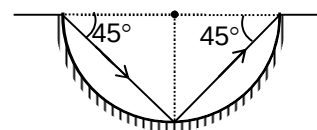


23. A hemispherical concave mirror is filled with a liquid of refractive index $\sqrt{2}$. A light enters into liquid at grazing incidence. The optical path of ray only due to medium at the emerging point is NR. Find the value of N. (R is radius of concave mirror)



Ans. 00004.00

Sol. Optical path = $n(2\sqrt{2}R) = 4R$



24. A radioactive sample contains two radioactive elements A and B having decay constant λ and 2λ . Initially 25% of total decay comes from A. After how many seconds 75% of total decay comes from A? (Take $\lambda = \ln 3$)

Ans. 00002.00

Sol. Ratio of initial activity

$$\frac{R_{A_0}}{R_{B_0}} = \frac{1}{3}$$

Ratio of final activity

$$\frac{R_{A_0} e^{-\lambda t}}{R_{B_0} e^{-2\lambda t}} = 3$$

$$\Rightarrow e^{\lambda t} = 9$$

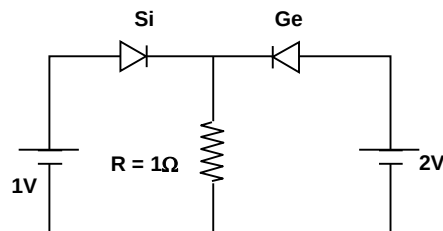
25. The k_α line wavelength is λ for an element with atomic number $z = 19$. Another atom having k_α wavelength $\left(\frac{\lambda}{4}\right)$ will have atomic number equal to

Ans. 00037.00

Sol.
$$\frac{\lambda_A}{\lambda_B} = \left(\frac{z_B - 1}{z_A - 1}\right)^2$$

$$z_B = 37$$

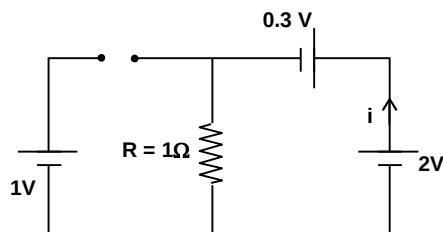
26. Two Silicon and Germanium diodes are connected in a circuit. The knee voltage of Si and Ge are 0.7 V and 0.3 V respectively. The current in resistance 1Ω is



Ans. 00001.70

Sol. As per knee voltage Ge is conducting but Si is blocking the current
 So current

$$i = \frac{2 - 0.3}{1} = 1.7 \text{ A}$$



27. Equation of an amplitude modulated wave is given as $C(t) = 20\cos(200\pi t) + 10\cos(200\pi t)\cos(2\pi t)$
 Then ratio of bandwidth and modulation index is :

Ans. 00001.00

Sol. $\omega_C = 200\pi \Rightarrow f_C = 100 \text{ Hz} \quad A_C = 20$

$$\omega_m = 2\pi \Rightarrow f_m = 1 \text{ Hz} \quad A_m = 20$$

$$\frac{\text{Band width}}{\mu} = \frac{2f_m}{A_C / A_m} = 1$$

28. A vernier calipers have 10 divisions on vernier scale coincide with 9 main scale division. When there is not any object between jaws and jaws touch each other the first main scale division coincides with first vernier scale division. If one main scale division is 1 mm then find distance moved by vernier calipers till third vernier scale division coincides with third main scale division (in mm).

Ans. 00000.20

Sol. 10 VSD = 9 MSD
 $\Rightarrow$ L.C. = 0.1 mm
 Distance = 0.1 mm $\times$ 2 = 0.2 mm

29. When water freezes into ice its volume increases by 2%. A rigid spherical shell is filled completely with water at 0°C. Now water is freezed completely then Bulk stress on surface of ice is $N \times 10^9$ Pa. Find the value of N. (Bulk's modulus of water = 2×10^{11} N/m²)

Ans. 00004.00

Sol. $B \frac{\Delta V}{V} = \Delta P$
 $0.02 \times 2 \times 10^{11} = \Delta P$
 $\Rightarrow \Delta P = 4 \times 10^9$

30. Two vectors $\vec{a}$ and $\vec{b}$ are such that
 $(\vec{a} - 2\vec{b}) \cdot (2\vec{a} + 3\vec{b}) = 2$
 and $(\vec{a} + \vec{b}) \cdot (2\vec{a} - \vec{b}) = 1$
 then the value of $4a^2 - 7b^2$ is

Ans. 00003.00

Sol. $2a^2 - 6b^2 - \vec{a} \cdot \vec{b} = 2$
 $2a^2 - b^2 + \vec{a} \cdot \vec{b} = 1$
 $4a^2 - 7b^2 = 3$

Chemistry

PART – B

SECTION – A (One Options Correct Type)

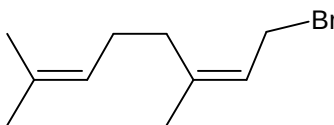
This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

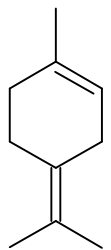
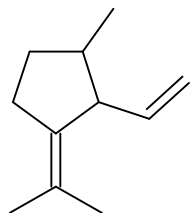
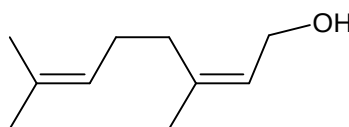
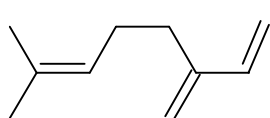
31. While calculating the strength of KMnO_4 using titration:

- (A) KMnO_4 is in burette and conc. H_2SO_4 is in conical flask.
- (B) KMnO_4 is in conical flask and conc. H_2SO_4 is in burette.
- (C) KMnO_4 is in burette and oxalic acid is in conical flask.
- (D) KMnO_4 is in conical flask and oxalic acid is in burette.

Ans. D

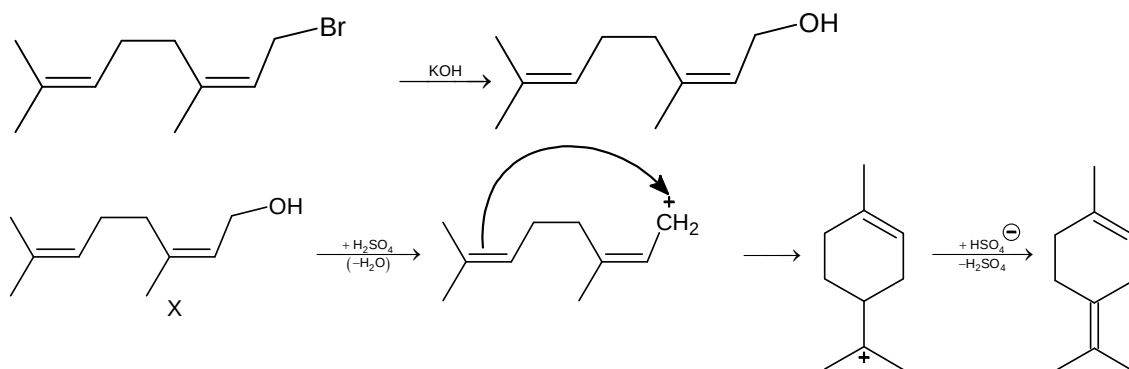
Sol. KMnO_4 is in conical flask and oxalic acid is added from burette. Cannot use H_2SO_4 as it is a secondary standard.

32.  $\xrightarrow{\text{KOH}}$ X $\xrightarrow{\text{conc. H}_2\text{SO}_4}$ Y; Y is :

- (A) 
- (B) 
- (C) 
- (D) 

Ans. A

Sol.

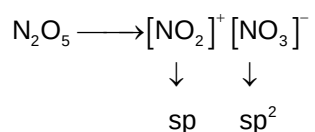


33. The hybridization of nitrogen atom in solid state structure of N_2O_5 is

- (A) sp^3, sp^3
- (B) sp^2, sp^2
- (C) sp, sp^2
- (D) sp^2, sp^3

Ans. C

Sol.



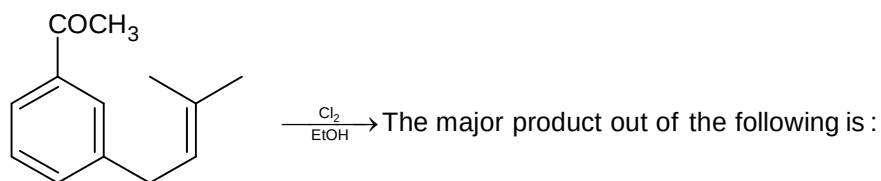
34. CFCl carbene and CCl_2 carbene are respectively:

- (A) Singlet, Singlet
- (B) Triplet, Triplet
- (C) Singlet, Triplet
- (D) Triplet, Singlet

Ans. A

Sol. Both CFCl and CCl_2 are capable of back-bonding, therefore, singlet state is more stable.

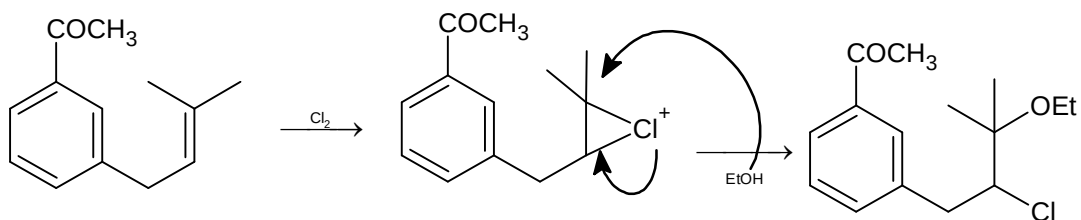
35.



- (A)
- (B)
- (C)
- (D)

Ans. C

Sol.

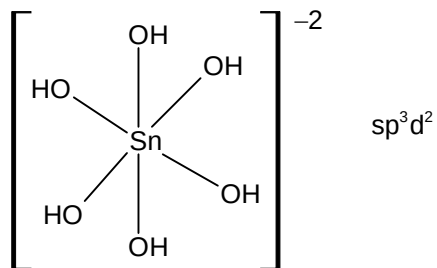


36. The hybridization of central atom in $[\text{H}_6\text{SnO}_6]^{2-}$ is:

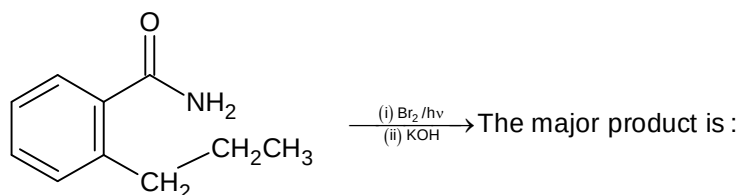
- (A) sp^3
 (B) sp^3d^2
 (C) sp^3d
 (D) sp^3d^3

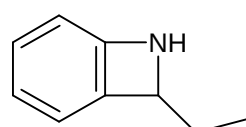
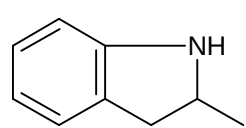
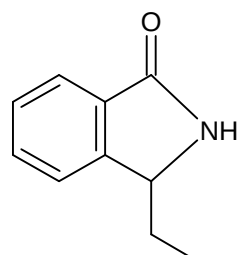
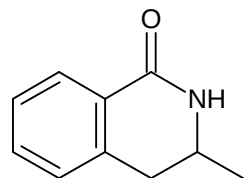
Ans. B

Sol.



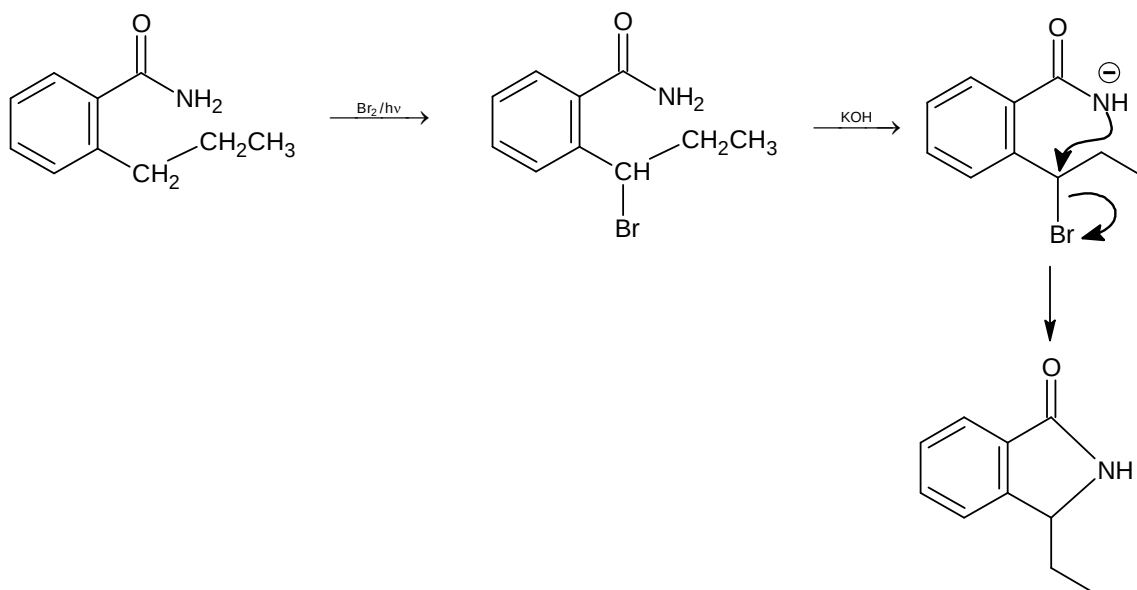
37.



- (A) 
- (B) 
- (C) 
- (D) 

Ans. C

Sol.

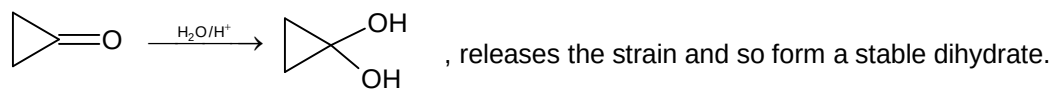


38. Which of the following form the most stable dihydrate?

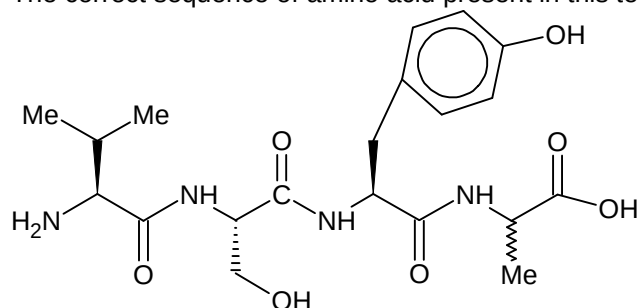
- (A)
- (B)
- (C)
- (D)

Ans. A

Sol.



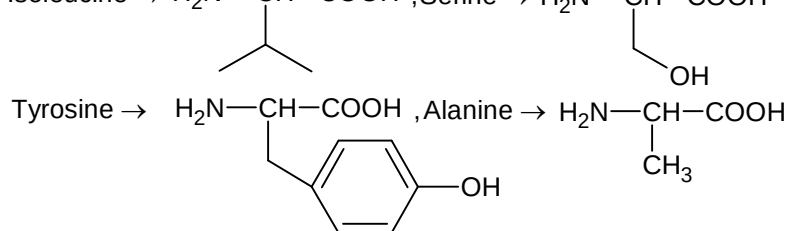
39. The correct sequence of amino acid present in this tetrapeptide given is:



- (A) Iso-Tyr-Ala-Ser
(B) Ala-Tyr-Iso-Ser
(C) Tyr-Ser-Iso-Ala
(D) Iso-Ser-Tyr-Ala

Ans. D

Sol. Isoleucine $\rightarrow$ $\text{H}_2\text{N}-\text{CH}-\text{COOH}$, Serine $\rightarrow$ $\text{H}_2\text{N}-\text{CH}-\text{COOH}$



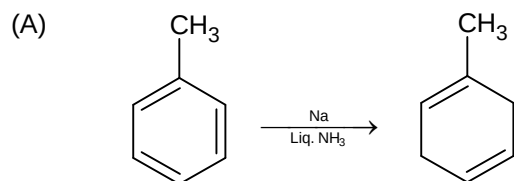
40. What types of forces are present in diethyl ether?

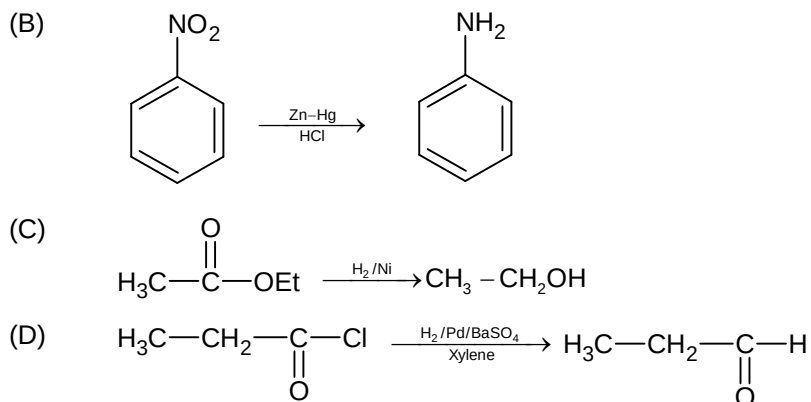
- (A) Hydrogen bonding only
(B) London dispersion forces and ion-dipole forces
(C) London dispersion forces and dipole-dipole forces
(D) Dipole-dipole forces and hydrogen bonding

Ans. C

Sol. Diethyl ether has slight polarity so, weak dipole-dipole forces exist along with London-dispersion forms.

41. Which of the following is an incorrect path of reduction under normal condition?

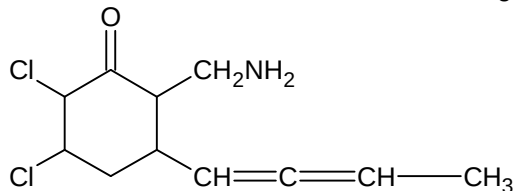




Ans. C

Sol. H_2 / Ni does not react with ester under normal conditions.

42. The total number of stereoisomers of the given compound is/are:



- (A) 8
(B) 16
(C) 64
(D) 32

Ans. D

Sol. 5 stereocenters are present.
Total isomer = $2^n = 2^5 = 32$.

43. For a reaction $A \longrightarrow P$, the energy of activation is 15.8 Kcal at 298 K. Calculate the number of activated molecules if we start with 1 mole of A, considering Arrhenius equation to be valid (Given: $e^{-26.51} = 3.067 \times 10^{-12}$)

- (A) 1.031×10^{21}
(B) 1.847×10^{12}
(C) 5.86×10^{23}
(D) 3.067×10^{12}

Ans. B

Sol.
$$\frac{\text{Total molecules activated}}{\text{Total number of molecules present}} = e^{-\frac{E_a}{RT}}$$

$$\frac{\text{Molecules activated}}{N_A (1 \text{ mole})} = e^{-\left(\frac{15.8 \times 10^3}{2 \times 298}\right)}$$

$$\text{Molecule activated} = 3.067 \times 10^{-12} \times 6.023 \times 10^{23}$$

$$= 1.847 \times 10^{12}$$

44. The flocculation values of these effective ions vary in the order for Gold sol as:

- (A) $\text{Sn}^{+4} < \text{Al}^{+3} < \text{Ba}^{+2} < \text{Na}^{+}$
- (B) $\text{PO}_4^{3-} < \text{SO}_4^{2-} < \text{Cl}^{-}$
- (C) $\text{Na}^{+} > \text{Al}^{+3} > \text{Ba}^{+2} > \text{Sn}^{+4}$
- (D) $\text{SO}_4^{2-} < \text{Na}^{+} < \text{PO}_4^{3-} < \text{Ba}^{+2}$

Ans. A

Sol. Flocculating value $\propto \frac{1}{\text{Flocculating power}}$

According to Hardy-Schulze rule: Flocculating power : $\text{Sn}^{+4} > \text{Al}^{+3} > \text{Ba}^{+2} > \text{Na}^{+}$

45. Equimolar solutions of KCl and compound X in water show depression in freezing point in the ratio of 4 : 1. Assuming KCl to be completely ionized, the compound X in the solutions must:

- (A) Dissociate to the extent of 50%
- (B) Hydrolyze to the extent of 80%
- (C) Dimerise to the extent of 50%
- (D) Trimerise to the extent of 75%

Ans. D

Sol. $m_{\text{KCl}} = m_X$

$$\therefore \frac{\Delta T_{f(\text{KCl})}}{i_{\text{KCl}}} = \frac{\Delta T_{f(X)}}{i_{(X)}}$$

$$i_{(X)} = \frac{1}{2}$$

For trimerisation to 75% extent

$$i = 1 + \left(\frac{1}{6} - 1\right) \times \frac{3}{4}$$

$$i = \frac{1}{2}$$

46. The process which is suitable for refining of Ti, Zr and Hf is

- (A) Zone refining
- (B) Mond's refining
- (C) Electrolytic refining
- (D) Van-Arkel method

Ans. D

Sol. All 3 belong to same group and Van-Arkel method is used for this refining.

47. If the colour of $[\text{Cr}(\text{H}_2\text{O})_6]\text{Cl}_3$ is violet, than the colour for $[\text{Cr}(\text{H}_2\text{O})_5\text{Cl}]\text{Cl}_2\text{H}_2\text{O}$ would be:

- (A) Yellow
- (B) Green
- (C) Violet
- (D) Red

Ans. B

Sol. $[\text{Cr}(\text{H}_2\text{O})_5\text{Cl}]\text{Cl}_2\text{H}_2\text{O}$ is green in colour which is expected as Cl is WFL with respect to H_2O hence λ_{abs} should increase.

48. NaAlSiO_4 is preferred for

- (A) Production of silicones
- (B) Removing permanent hardness of water
- (C) Calculating strength of KMnO_4
- (D) Manufacturing high quality soaps

Ans. B

Sol. NaAlSiO_4 is a Zeolite used for removing permanent hardness of water.

49. Acid rain can have a pH value of:

- (A) 4.6
- (B) 5.7
- (C) 6.6
- (D) 7.0

Ans. A

Sol. Acid rain has pH < 5.6.

50. Select the incorrect statement:

- (A) Novalac is linear polymer
- (B) Glyptal is a polymer of ethylene glycol and phthalic acid
- (C) Bakelite is a cross-linked phenol formaldehyde polymer
- (D) In manufacturing of tyre rubber, 10% of sulphur is used as a cross-linking agent

Ans. D

Sol. 5% of sulphur is used for manufacturing of tyre rubber

SECTION – B (Numerical Answer Type)

This section contains **10** questions. The answer to each question is a **NUMERICAL VALUE**. For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

51. The EMF of the cell $M | M^{n+} (0.02 \text{ M}) || H^+ (1 \text{ M}) | H_2 (g) (1 \text{ atm})$ at 25°C is 0.81 V. The valency of metal if the standard oxidation potential of $M | M^{n+}$ is 0.76 V, is:

Ans. 00002.00

Sol. $M(s) + nH^+ \longrightarrow M^{n+}(aq) + \frac{n}{2}H_2(g)$

$$E_{\text{cell}} = E_{\text{cell}}^{\circ} - \frac{0.059}{n} \log \frac{[M^{n+}](P_{H_2})^{n/2}}{[H^+]^n}$$

$$0.81 - 0.76 = \frac{-0.059}{n} \log(0.02)$$

$$n = 2$$

52. Calculate the entropy change involved to convert 2 Kg $H_2O(g)$ at 385 K to $H_2O(l)$ at 280 K in $(\text{kJ kg}^{-1} \text{K}^{-1})$, (magnitude only).

Given: specific heat of water liquid and water vapour are 4.2 and 2 $\text{kJ K}^{-1} \text{kg}^{-1}$ respectively. Latent heat of vaporization is 2491 $\text{kJ K}^{-1} \text{kg}^{-1}$. [Take $\log 385 = 2.58$, $\log 373 = 2.57$, $\log 280 = 2.45$]

Ans. 00009.20

Sol. $H_2O(g) \xrightarrow{385 \text{ K}} H_2O(g) \rightleftharpoons H_2O(l) \xrightarrow{373 \text{ K}} H_2O(l) \xrightarrow{280 \text{ K}}$

$$\Delta S = mC \ln \frac{T_2}{T_1}$$

$$\Delta S_1 = 2 \times 2 \times 2.303 \log \left(\frac{373}{385} \right) = -0.1267$$

$$\Delta S_2 = \frac{\Delta H_{\text{condensation}}}{T} = -\frac{2491}{373} = -6.67$$

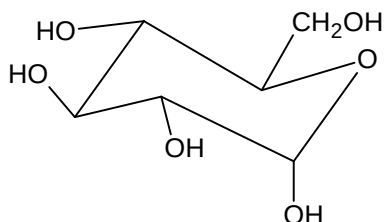
$$\Delta S_3 = 2 \times 4.2 \times 2.303 \log \left(\frac{280}{385} \right) = -2.409$$

$$\Delta S_{\text{Total}} = -9.205 \text{ kJ kg}^{-1} \text{ K}^{-1}$$

53. The number of –OH groups acquiring equatorial position in most stable form of α -D-(+)-Glucopyranose are:

Ans. 00003.00

Sol.



Only 3 are equatorial position.

54. The ideal limiting radius ratio of $[\text{Ni}(\text{CO})_4]$ is:

Ans. 00000.22

Sol. $[\text{Ni}(\text{CO})_4]$ is sp^3 hybridised, hence tetrahedral.

Limiting ratio for tetrahedral is 0.225.

55. Given $\psi_{2s} = \left(\frac{Z}{2a_0} \right)^{1/2} \left(2 - \frac{Zr}{a_0} \right) e^{-Zr/a_0}$, the distance from the nucleus of the radial node of 2s electron of Li^{+2} ion is equal to xa_0 (a_0 = Bohr radius). The value of x is:

Ans. 00000.67

Sol. For node, $\psi_{2s}^2 = 0 \therefore \psi_{2s} = 0$

$$\left(2 - \frac{Zr}{a_0} \right) = 0$$

$$r = \frac{2a_0}{Z} = \frac{2a_0}{3} = 0.67a_0$$

56. 100 mL sample of tap water contain 18 mg MgSO_4 . If the hardness of sample (in terms of CaCO_3) is x ppm. Calculate $\frac{x}{8}$?

Ans. 00018.75

Sol. 100 mL $\text{H}_2\text{O} \longrightarrow 18 \text{ mg } \text{MgSO}_4$

$$\therefore 1 \text{ L} \longrightarrow 180 \text{ mg MgSO}_4$$

$$n_{\text{MgSO}_4} = n_{\text{CaCO}_3} = \frac{180 \text{ mg}}{120 \text{ g/mol}}$$

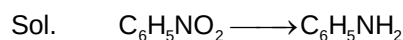
$$w_{\text{CaCO}_3} = 150 \text{ mg in 1 L}$$

$$\therefore x = 150 \text{ ppm}$$

$$\therefore \frac{x}{8} = \frac{150}{8} = 18.75$$

57. Calculate P, if the amount of charge required (in terms of Faraday's constant) is PF to reduce 12.3 g of nitrobenzene completely in acidic medium.

Ans. 00000.60



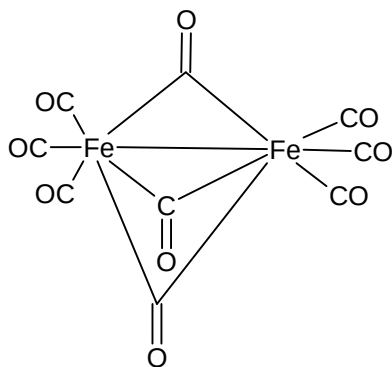
$$Q = n \times nf \times F$$

$$= \frac{12.3}{123} \times 6 \times F$$

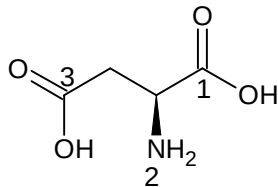
58. The number of metal-metal bond in $[\text{Fe}_2(\text{CO})_9]$ is/are:

Ans. 00001.00

Sol.



59. Calculate the isoelectric point for Aspartic acid



$$\text{pK}_{a_1}(\text{COOH}) = 1.88$$

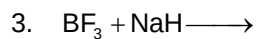
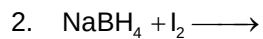
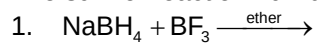
$$\text{pK}_{a_2}(\text{NH}_2) = 9.60$$

$$\text{pK}_{a_3}(\text{COOH}) = 3.65$$

Ans. 00002.76

Sol.
$$\text{pI} = \frac{\text{pK}_{a_1} + \text{pK}_{a_3}}{2} = \frac{1.88 + 3.65}{2} = 2.765$$

60. The sum of reaction numbers that can be used for preparation of diborane amongst given is



Ans. 00006.00

Sol. All three reactions can be used.

$\therefore$ Sum is $(1 + 2 + 3) = 6$.

Mathematics

PART – C

SECTION – A (One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

61. If $r \geq 40$, then the greatest possible value of ${}^{40}C_0 \cdot {}^{60}C_r + {}^{40}C_1 \cdot {}^{60}C_{r-1} + {}^{40}C_2 \cdot {}^{60}C_{r-2} \dots \dots {}^{40}C_{40} \cdot {}^{60}C_{r-40}$ is

- (A) ${}^{200}C_{50}$
(B) ${}^{60}C_{50}$
(C) ${}^{100}C_{50}$
(D) ${}^{200}C_{100}$

Ans. C

Sol. ${}^{40}C_0 \cdot {}^{60}C_r + {}^{40}C_1 \cdot {}^{60}C_{r-1} + {}^{40}C_2 \cdot {}^{60}C_{r-2} \dots \dots {}^{40}C_{40} \cdot {}^{60}C_{r-40} = {}^{100}C_r$
This will be maximum when $r = 50$

62. The value of k for which the equation $|x^2 - 5x + 6| = k$ has three real and distinct roots is

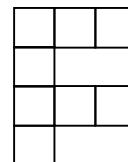
- (A) 4
(B) 2
(C) $\frac{1}{4}$
(D) $\frac{1}{2}$

Ans. C

Sol. $|x^2 - 5x + 6| = k$ has three real roots when the line $y = k$ touches the parabola $y = -(x^2 - 5x + 6)$ at vertex

$$\Rightarrow k = \left| -\frac{D}{4a} \right| = \left| -\frac{1}{4} \right| = \frac{1}{4}$$

63. The number of ways in which letters of word IITJEE can be placed in the squares of the figure as shown, so that no row remains empty, are



- (A) 1620
(B) 810
(C) 5040
(D) 2520

Ans. A

Sol. Arrangement of IITJEE is $\frac{6!}{2!2!} = 180$ ways 6 place can be selected by ${}^3C_2 \cdot {}^3C_2$ ways
Total ways = $9 \times 180 = 1620$

64. Let O be the origin and $\overrightarrow{OA} = 3\hat{i} - 2\hat{j} + \hat{k}$, $\overrightarrow{OB} = \hat{i} + \hat{j} - 2\hat{k}$ and $\overrightarrow{OC} = \alpha\overrightarrow{OB} + \beta\overrightarrow{OA}$ for some $\alpha, \beta > 0$.
If $|\overrightarrow{OB} \times \overrightarrow{OC}| = \sqrt{83}$ and $|\overrightarrow{OC}| = \sqrt{14}$, then

(A) $\alpha = 1, \beta = \frac{1}{3}$

(B) $\alpha = \frac{1}{3}, \beta = 2$

(C) $\alpha = \frac{1}{3}, \beta = 1$

(D) $\alpha = \frac{1}{3}, \beta = \frac{2}{3}$

Ans. C

Sol. $|\overrightarrow{OB} \times \overrightarrow{OC}| = |\alpha(\overrightarrow{OB} \times \overrightarrow{OB}) + \beta(\overrightarrow{OB} \times \overrightarrow{OA})| = \sqrt{83}$
 $\Rightarrow |\beta(-3\hat{i} - 7\hat{j} - 5\hat{k})| = \sqrt{83} \Rightarrow \beta = 1$
 Now, $\overrightarrow{OC} = \alpha(\hat{i} + \hat{j} - 2\hat{k}) + (3\hat{i} - 2\hat{j} + \hat{k})$
 $|\overrightarrow{OC}| = |(\alpha + 3)\hat{i} + (\alpha - 2)\hat{j} + (-2\alpha + 1)\hat{k}| = \sqrt{14}$
 $\alpha = \frac{1}{3}$

65. Let E, F and G be three events, probability of event $P(E \cap F) = \frac{1}{2}$, $P(E \cap G) = \frac{1}{3}$,
 $P(E \cap F \cap G) = \frac{1}{6}$. For any event A, if A^c denotes its complement, then $P(E \cap F \cap G^c) + P(E \cap G \cap F^c)$ is

(A) $\frac{1}{3}$

(B) $\frac{1}{6}$

(C) $\frac{1}{2}$

(D) $\frac{2}{3}$

Ans. C

Sol. $P(E \cap F \cap G^c) = P(E \cap F) - P(E \cap F \cap G)$
 $P(E \cap G \cap F^c) = P(E \cap G) - P(E \cap F \cap G)$

66. The number of 3×3 matrices M with entries from elements of set $\{0, 1, 2, 3, 4\}$ are there for which the $\text{tr}(M) = 5$ are (where $\text{tr}(M)$ denotes trace of matrix M)

- (A) $12(5^6)$
- (B) $6(5^6)$
- (C) $21(5^6)$
- (D) $18(5^6)$

Ans. D

Sol. Let diagonal elements are x_1, x_2, x_3 , then $x_1 + x_2 + x_3 = 5$
 $\Rightarrow$ Coefficient of x^5 in $(x^0 + x^1 + x^2 + x^3 + x^4)^3$
 $\Rightarrow 18$
 Now, total matrices are $18(5^6)$

$$\begin{bmatrix} x_1 & - & - \\ - & x_2 & - \\ - & - & x_3 \end{bmatrix}$$

67. A debate club consists 6 girls and 4 boys. A team of 4 member is to be selected from this club including the selection of a captain and a vice captain (from among these 4 members) for the team. If the team has to include atmost two boy, then the number of way of selecting the team is

- (A) 2200
- (B) 2220
- (C) 380
- (D) 2320

Ans. B

Sol. Number of way of selection of 4 girls = 6C_4
 Number of way of selection of 3 girls, one boy = ${}^6C_3 \cdot {}^4C_1$
 Number of way of selection of 2 girls and 2 boys = ${}^6C_2 \cdot {}^4C_2$
 Now, number of way of selection of one captain and one vice captain = ${}^4C_2 \cdot 2!$
 Total way = $({}^6C_4 + {}^6C_3 \cdot {}^4C_1 + {}^6C_2 \cdot {}^4C_2) {}^4C_2 \times 2!$

68. The value of b for which the common tangent to ellipse $\frac{x^2}{4} + \frac{y^2}{b^2} = 1$ and the parabola $y^2 = 4x$ and their chord of contact to parabola form an equilateral triangle

- (A) $\sqrt{\frac{2}{3}}$
- (B) $\sqrt{\frac{5}{3}}$
- (C) $\sqrt{\frac{7}{3}}$
- (D) $\sqrt{\frac{11}{3}}$

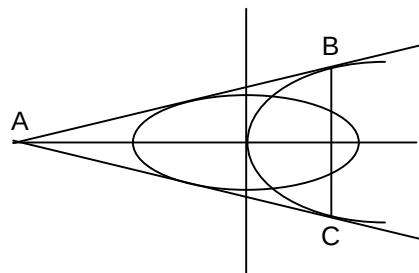
Ans. B

Sol. Let common tangent AB and AC, intersect on x-axis
(by symmetry)

$\Rightarrow \triangle ABC$ is equilateral if $\angle BAC = 60^\circ$

$\Rightarrow$ Slope of AB = $\frac{1}{\sqrt{3}}$; Slope of AC = $-\frac{1}{\sqrt{3}}$

$\Rightarrow \frac{1}{m} = \sqrt{4m^2 + b^2} \Rightarrow b^2 = \frac{5}{3}$



69. If $f(x)$ be a differentiable function such that $f'(2) = 1$, $f'(3) = 2$, then $\lim_{h \rightarrow 0} \frac{f(2+5h^2-3h) - f(2)}{f(3+h^2-2h) - f(3)}$ equal to

(A) $\frac{1}{4}$

(B) $\frac{1}{2}$

(C) $-\frac{3}{2}$

(D) $\frac{3}{4}$

Ans. D

Sol. $\lim_{h \rightarrow 0} \frac{f(2+5h^2-3h) - f(2)}{f(3+h^2-2h) - f(3)}$ by L. Hospital Rule

$$= \lim_{h \rightarrow 0} \frac{f'(2+5h^2-3h)(10h-3)}{f'(3+h^2-2h)(2h-2)} = \frac{3}{4}$$

70. Equation of common tangent of parabola $y^2 = 12x$ and $y^2 = 16(x-3)$ is

(A) $\sqrt{3}x - y + 3 = 0$

(B) $\sqrt{3}y - x - 9 = 0$

(C) $x - y + \sqrt{3} = 0$

(D) $\sqrt{3}x - 3y + 3 = 0$

Ans. B

Sol. Equation of common tangent to both parabola

$y^2 = 12x$ equation of tangent $y = mx + \frac{3}{m}$

$$y^2 = 16(x-3); y = m(x-3) + \frac{4}{m} \text{ by comparing both slope} = \pm \frac{1}{\sqrt{3}}$$

$$\text{So, equation of common tangent } y = \pm \frac{1}{\sqrt{3}}x + \frac{3}{\left(\pm \frac{1}{\sqrt{3}}\right)}$$

71. The point on the curve $x + y^2 = 0$ which is closest to the point $(0, -3)$

- (A) $\left(-2, 2^{\frac{1}{2}}\right)$
- (B) $(-1, -1)$
- (C) $(-1, 1)$
- (D) $\left(-2, -2^{\frac{1}{2}}\right)$

Ans. B

Sol. For parabola $y^2 = -x$ equation of normal $\Rightarrow y = mx + \frac{1}{2}m + \frac{1}{4}m^3$ this passes through $(0, -3)$

$$-3 = m(0) + \frac{1}{2}m + \frac{1}{4}m^3 \Rightarrow m = -2$$

$$\text{So, point will be } (-a(-m)^2, -2a(-m)) = \left(-\frac{1}{4}(-2)^2, -2\left(\frac{1}{4}\right) \cdot 2\right) = (-1, -1)$$

72. Let a, b, c be the sides, m_a, m_b, m_c the lengths of the altitudes and d_a, d_b, d_c the distances from the vertices to the orthocentre in an acute triangle, then $m_a d_a + m_b d_b + m_c d_c$ is equal to

- (A) $a^2 + b^2 + c^2$
- (B) $2(a^2 + b^2 + c^2)$
- (C) $\frac{a^2 + b^2 + c^2}{2}$
- (D) $ab + bc + ca$

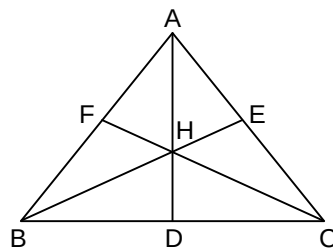
Ans. C

Sol. Let D, E, F are altitudes from A, B, C respectively H be the orthocentre of $\triangle ABC$, then the $\triangle ACD$ and $\triangle HAE$ are similar triangle

So, $AD \cdot AH = m_a \cdot d_a = AC \cdot AE = AE \cdot b$ similarly in $\triangle ADB$ and $\triangle AHF$
 $m_a \cdot d_a = AF \cdot c$

$$\text{So, } m_a \cdot d_a = \frac{AF \cdot c + AE \cdot b}{2}. \text{ Similarly } m_b \cdot d_b = \frac{BF \cdot c + BD \cdot a}{2},$$

$$m_c d_c = \frac{CD \cdot a + CE \cdot b}{2}$$



Therefore $m_a d_a + m_b d_b + m_c d_c$

$$= \frac{1}{2} \{ AE \cdot b + AF \cdot c + BF \cdot c + BD \cdot a + CD \cdot a + CE \cdot b \}$$

$$= \frac{1}{2} \{ (BD + CD) \cdot a + (AE + CE) \cdot b + (BF + AF) \cdot c \} = \frac{1}{2} (a^2 + b^2 + c^2)$$

73. Let consider a function defined in R , $f(x) = \frac{1}{(1+e^x)^2}$, then which of the following option is correct?

- (A) $\lim_{x \rightarrow 0^+} f(x) = \frac{1}{2}$
- (B) $f(x)$ is discontinuous at exactly one point in R
- (C) $f(x)$ is non differentiable at $x = 0$
- (D) $x = -\ln 2$ is point of inflection

Ans. D

Sol. $f(x) = \frac{1}{(1+e^x)^2}$ is always continuous and differentiable in R

$$f'(x) = -\frac{2e^x}{(1+e^x)^3}$$

$$f''(x) = \frac{2e^x(2e^x - 1)}{(1+e^x)^4} \text{ clearly } x = -\ln 2 \text{ is point of inflection}$$

74. The contrapositive of $p \rightarrow (\sim q \rightarrow \sim r)$ is

- (A) $(\sim q \wedge r) \rightarrow \sim p$
- (B) $(q \rightarrow r) \rightarrow \sim p$
- (C) $(q \vee \sim r) \rightarrow \sim p$
- (D) $(\sim q \vee \sim r) \rightarrow \sim p$

Ans. A

Sol. Contrapositive will be $\sim(\sim q \rightarrow \sim r) \rightarrow \sim p$

$$\Rightarrow \sim(q \vee \sim r) \rightarrow \sim p$$

$$\Rightarrow (\sim q \wedge r) \rightarrow \sim p$$

75. The mean square deviation of a set of n observations $x_1, x_2, \dots, x_n$ about a point c is defined as $\frac{1}{n} \sum_{i=1}^n (x_i - c)^2$. The mean square deviation about -2 and 2 are 18 and 10 respectively, then the standard deviation of this set of observation is

- (A) 3

- (B) 2
(C) 1
(D) 4

Ans. A

Sol. $\frac{1}{n} \sum_{i=1}^n (x_i - 2)^2 = 10$ and $\frac{1}{n} \sum_{i=1}^n (x_i + 2)^2 = 18$

$$\Rightarrow \sum_{i=1}^n (x_i + 2)^2 + \sum_{i=1}^n (x_i - 2)^2 = 28n \text{ and } \sum_{i=1}^n (x_i + 2)^2 - \sum_{i=1}^n (x_i - 2)^2 = 8n$$

$$\Rightarrow 2 \sum_{i=1}^n x_i^2 + 8n = 28n \text{ and } 8 \sum_{i=1}^n x_i = 8n \Rightarrow \sum_{i=1}^n x_i^2 = 10n \text{ and } \sum_{i=1}^n x_i = n$$

$$\sigma = \sqrt{\frac{\sum_{i=1}^n x_i^2}{n} - \left(\frac{\sum_{i=1}^n x_i}{n}\right)^2} = \sqrt{\frac{10n}{n} - \left(\frac{n}{n}\right)^2} = 3$$

76. The value of the integral $I = \int_2^3 \sqrt{(x-2)(3-x)} dx$ is equal to

- (A) $\frac{\pi}{4}$
(B) $\frac{\pi}{2}$
(C) $\frac{\pi}{8}$
(D) 2π

Ans. C

Sol. $I = \int_2^3 \sqrt{(x-2)(3-x)} dx$ put $x = 2 \cos^2 \theta + 3 \sin^2 \theta$

$$I = \frac{\pi}{8}$$

77. Let $f : [0, \infty) \rightarrow \mathbb{R}$ be a continuous and strictly increasing function such that $f^4(x) = \int_0^x t \cdot f^3(t) dt \forall x > 0$. The area enclosed by $y = f(x)$, x -axis and the ordinate at $x = 2\sqrt{2}$ is

- (A) $2\sqrt{2}$
(B) $\frac{2\sqrt{2}}{5}$

(C) $\frac{2\sqrt{2}}{3}$

(D) $\frac{8}{3}$

Ans. C

Sol. $f^4(x) = \int_0^x t \cdot f^3(t) dt \Rightarrow f'(x) = \frac{x}{4} \Rightarrow f(x) = \frac{x^2}{8} + c, f(0) = 0$

$\Rightarrow c = 0; f(x) = \frac{x^2}{8}$

Area = $\int_0^{2\sqrt{2}} \frac{x^2}{8} dx = \left[\frac{x^3}{24} \right]_0^{2\sqrt{2}} = \frac{2\sqrt{2}}{3}$

78. Tangents are drawn from any point of the hyperbola $\frac{x^2}{9} - \frac{y^2}{4} = 1$ to the circle $x^2 + y^2 = 1$, then the locus of midpoint of chord of contact is

(A) $\frac{x^2}{2} - \frac{y^2}{3} = (x^2 + y^2)^2$

(B) $\frac{x^2}{9} - \frac{y^2}{2} = 3(x^2 + y^2)^2$

(C) $\frac{x^2}{9} + \frac{y^2}{4} = 3(x^2 + y^2)^2$

(D) $\frac{x^2}{9} - \frac{y^2}{4} = (x^2 + y^2)^2$

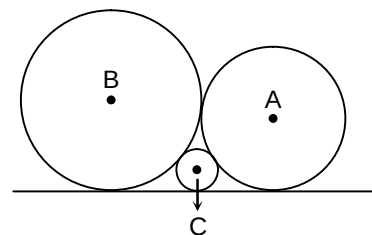
Ans. D

Sol. Let any point on hyperbola is $(3 \sec \theta, 2 \tan \theta)$
Equation of chord of contact $(3 \sec \theta)x + (2 \tan \theta)y = 1$ (1)
Also, let midpoint (h, k) of chord of contact so equation of chord
 $xh + yk = h^2 + k^2$ (2)

Equation (1) and (2) are equal $\frac{3 \sec \theta}{h} = \frac{2 \tan \theta}{k} = \frac{1}{h^2 + k^2}$

$\Rightarrow \sec^2 \theta - \tan^2 \theta = 1 \Rightarrow \frac{h^2}{9(h^2 + k^2)^2} - \frac{k^2}{4(h^2 + k^2)^2} = 1$

79. Three circles are tangent to each other and to the straight line. The radii of the circles with centres A, B and C are 4, 9, r respectively, then r is equal to



- (A) 1.42
(B) 1.43
(C) 1.44
(D) 1.45

Ans. C

Sol. If two circle touches each other externally, then

$$L_{\text{ext}} = 2\sqrt{r_1 r_2} \text{ when } r_1 \text{ and } r_2 \text{ are radii of circles}$$

$$\Rightarrow 2\sqrt{4 \cdot 9} = 2\sqrt{r \cdot 4} + 2\sqrt{r \cdot 9}$$

$$\Rightarrow r = \frac{36}{25}$$

80. Orthogonal trajectories of family of the curve $x^{\frac{2}{3}} + y^{\frac{2}{3}} = a$ where a is an arbitrary constant, is

- (A) $x^{\frac{2}{3}} - y^{\frac{2}{3}} = c$
(B) $x^{\frac{2}{3}} + y^{\frac{2}{3}} = cx$
(C) $x^{\frac{4}{3}} - y^{\frac{4}{3}} = c$
(D) $x^{\frac{4}{3}} - y^{\frac{4}{3}} = cx$

Ans. C

Sol. $x^{\frac{2}{3}} + y^{\frac{2}{3}} = a$

$$\text{Differential equation : } x^{-\frac{1}{3}} + y^{-\frac{1}{3}} \frac{dy}{dx} = 0$$

$$\text{Orthogonal differential equation : } x^{-\frac{1}{3}} - y^{-\frac{1}{3}} \frac{dx}{dy} = 0$$

$$\text{Curve } x^{\frac{4}{3}} - y^{\frac{4}{3}} = c$$

SECTION – B
(Numerical Answer Type)

This section contains **10** questions. The answer to each question is a **NUMERICAL VALUE**. For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

81. If z lies on locus $\arg(z - 1 - 2i) = \frac{\pi}{4}$ and $\arg(z - 1 + 2i) = \frac{\pi}{6}$, then the number of complex number which satisfy both the locus

Ans. 00000.00

Sol. As both of ray

$\arg(z - 1 - 2i) = \frac{\pi}{4}$ the originate ray (1) from $z_1 \equiv (1 + 2i)$ and having argument $\frac{\pi}{4}$ and ray (2)

$\arg(z - 1 + 2i) = \frac{\pi}{6}$ is originating from $(1, -2)$ having argument $\frac{\pi}{6}$

So, $\arg(\text{ray 1}) > \arg(\text{ray 2})$ they will never intersect

82. If z lies on locus $||z - 2| - |z + 2|| = 2$, then the $\min|z - 4|$ is

Ans. 00003.00

Sol. z lies on hyperbola $\frac{x^2}{1} - \frac{y^2}{3} = 1$

83. If $bx^5 + cx^4 + dx^3 + ex^2 + fx + g = \begin{vmatrix} x^3 - 1 & 2x^2 \\ 4x^3 & 5x \end{vmatrix}$ is an identity in x , where b, c, d, e, f, g are constants, then $|f|$ is equal to

Ans. 00005.00

Sol. Differentiate once both side and put $x = 0$

$$\begin{vmatrix} 3x^2 & 4x \\ 4x^3 & 5x \end{vmatrix} + \begin{vmatrix} x^3 - 1 & 2x^2 \\ 12x^2 & 5 \end{vmatrix} = 5bx^4 + 4cx^4 + 3dx^3 + 2ex + f$$

$$\Rightarrow f = -5$$

84. Let $S = \{(i, j, k) ; 1 \leq i < j \leq k \leq 20 ; i, j, k, \in \{1, 2, \dots, 20\}\}$, then total number of elements in set S are

Ans. 01330.00

Sol. **Case-1:** $1 \leq i < j < k \leq 20 = {}^{20}C_3$

Case-2: $1 < i < j = k \leq 20 = {}^{20}C_2$

So, total elements $= {}^{20}C_3 + {}^{20}C_2 = {}^{21}C_3 = 1330$

85. The value of ${}^{100}C_0 + {}^{100}C_3 + {}^{100}C_6 + {}^{100}C_9 \dots {}^{100}C_{99}$ is $\frac{2^{k_1} - 1}{k_2}$, then $k_1 - k_2$ is equal to

Ans. 00097.00

Sol. $(1+x)^{100} = {}^{100}C_0 + {}^{100}C_1x^1 + {}^{100}C_2x^2 + \dots + {}^{100}C_{100}x^{100}$
 $(1+1)^{100} = {}^{100}C_0 + {}^{100}C_1 + {}^{100}C_2 + {}^{100}C_3 + \dots + {}^{100}C_{100}$
 $(1+\omega)^{100} = {}^{100}C_0 + {}^{100}C_1\omega + {}^{100}C_2\omega^2 + \dots + {}^{100}C_{100}\omega$
 $(1+\omega^2)^{100} = {}^{100}C_0 + {}^{100}C_1\omega^2 + {}^{100}C_2\omega + \dots + {}^{100}C_{100}\omega^2$
 $2^{100} + \omega^{200} + \omega^{100} = 3 \{ {}^{100}C_0 + {}^{100}C_3 + {}^{100}C_6 + {}^{100}C_9 + \dots + {}^{100}C_{99} \}$
 $\frac{2^{100}-1}{3} = {}^{100}C_0 + {}^{100}C_3 + {}^{100}C_6 + {}^{100}C_9 + \dots + {}^{100}C_{99}$

86. Suppose that $f : \mathbb{R} \rightarrow \mathbb{R}$ is continuous and satisfies the equation $f(x) \cdot f(f(x)) = 1$ for all real x . Assuming that $f(1000) = 999$, then the value of $f(10)$ is

Ans. 00000.10

Sol. We have $f(1000) = 999$; $f(1000) \cdot f(f(1000)) = 1$; $f(f(1000)) = \frac{1}{999}$

So, the range of f contains $\frac{1}{999}$ and 999 so by intermediate mean value theorem. Let $f(a) = 10$

So, $f(a) \cdot f(f(a)) = 1$; $10f(10) = 1$; $f(10) = \frac{1}{10} = 0.1$

87. If $a > 0$, $p \geq 2$ evaluate the value of $\lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{1}{\sqrt[p]{n^p + ra}}$

Ans. 00001.00

Sol. By Squeeze Play Theorem

$$\frac{1}{\sqrt[p]{n^p + na}} \leq \frac{1}{\sqrt[p]{n^p + ra}} \leq \frac{1}{\sqrt[p]{n^p + a}}, \text{ then } n \cdot \frac{1}{\sqrt[p]{n^p + na}} \leq \sum_{r=1}^n \frac{1}{\sqrt[p]{n^p + ra}} \leq \frac{n}{\sqrt[p]{n^p + a}}$$

$$\lim_{n \rightarrow \infty} \frac{n}{\sqrt[p]{n^p + na}} = \lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{n}{\sqrt[p]{n^p + a}} = 1 = \lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{1}{\sqrt[p]{n^p + ra}}$$

88. If $\lim_{n \rightarrow \infty} \sum_{k=1}^{n+1} \tan^{-1} \left(\frac{1}{2k^2} \right) = \lambda$, then $[\lambda]$ is equal to (where $[.]$ denotes greatest integer function)

Ans. 00000.00

Sol. $\tan^{-1} \left(\frac{1}{2k^2} \right) = \tan^{-1} \left(\frac{k}{k+1} \right) - \tan^{-1} \left(\frac{k-1}{k} \right) \Rightarrow \lim_{n \rightarrow \infty} \sum_{k=1}^{n+1} \tan^{-1} \left(\frac{1}{2k^2} \right) = \lim_{n \rightarrow \infty} \tan^{-1} \left(\frac{n}{n+1} \right) = \frac{\pi}{4}$

89. A rectangle with sides parallel to the coordinate axes is to be inscribed in the region enclosed by the graphs of $y = x^2$ and $y = 4$, then maximum length of its perimeter is _____ (in unit)

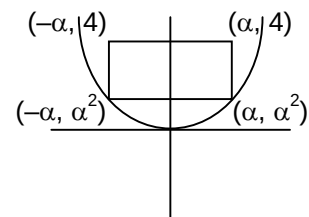
Ans. 00010.00

Sol. Perimeter = $2(2\alpha) + 2(4 - \alpha^2)$

$$P = 4\alpha + 8 - 2\alpha^2 \text{ extrema when } \frac{dP}{d\alpha} = 0 = 4 - 4\alpha \Rightarrow \alpha = 1$$

$$\frac{d^2P}{d\alpha^2} = -4 < 0 \text{ local maxima at } \alpha = 1$$

$$\text{So, perimeter } P = 4\alpha + 8 - 2\alpha^2 \Rightarrow 4 + 8 - 2 = 10$$



90. Suppose that a, b, c, d are positive real numbers satisfying $(a + c)(b + d) = ac + bd$, then the smallest possible value of $\frac{a}{b} + \frac{b}{c} + \frac{c}{d} + \frac{d}{a}$ is

Ans. 00008.00

Sol. Apply A.M. $\geq$ G.M.

$$\left(\frac{a}{b} + \frac{c}{d}\right) + \left(\frac{b}{c} + \frac{d}{a}\right) \geq 2 \left\{ \sqrt{\frac{ac}{bd}} + \sqrt{\frac{bd}{ac}} \right\}$$

$$\geq 2 \left\{ \frac{ac + bd}{\sqrt{abcd}} \right\}$$

$$\geq 2 \left\{ \frac{(a + c)(b + d)}{\sqrt{abcd}} \right\}$$

$$\geq 2 \left\{ \frac{2\sqrt{ac} \cdot 2\sqrt{bd}}{\sqrt{abcd}} \right\}$$

$$\geq 8$$